

Step By Step SELinux test on terminal & Apache

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After the introduction and the SELinux principle, it's time for the commands on Linux terminal

Step 1: Install and Start httpd (Apache) to test the difference between enabling and disabling SELinux

- 1- Install Apache `sudo yum install httpd -y`
- 2- Start and enable the Apache service `sudo systemctl start httpd`
`sudo systemctl enable httpd`

Step 2: Set up Two Servers as testing ground

- Server 1: SELinux-enabled

- 1-creating the server `sudo mkdir -p /var/www/html/secure`
- 2-Add a test file `echo "This is the SELinux-enabled server" | sudo tee /var/www/html/secure/index.html`
- 3- Set the correct SELinux context for the directory and file `sudo chcon -R -t httpd_sys_content_t /var/www/html/secure`
- 4- Configure Apache to serve this directory on port 8080: Edit or create `sudo nano /etc/httpd/conf.d/secure.conf`
In Apache window that will pop out write the following:

```
<VirtualHost *:8080>
    DocumentRoot "/var/www/html/secure"
    <Directory "/var/www/html/secure">
        Require all granted
    </Directory>
</VirtualHost>
```

- Server 2: SELinux-disabled

- 1-creating the server `sudo mkdir -p /var/www/html/insecure`
- 2-Add a test file `echo "This is the SELinux-disabled server" | sudo tee /var/www/html/insecure/index.html`
- 3- Disable SELinux enforcement temporarily: `sudo setenforce 0`
- 4- Configure Apache to serve this directory on port 8081: Edit or create `sudo nano /etc/httpd/conf.d/insecure.conf`
In Apache window that will pop out write the following:

```
<VirtualHost *:8081>
    DocumentRoot "/var/www/html/insecure"
    <Directory "/var/www/html/insecure">
        Require all granted
    </Directory>
</VirtualHost>
```

- ◆ After creating the two servers, now it's time to restart Apache to apply changes:
`sudo systemctl restart httpd`

Step 3: Test Both Servers

- 1-SELinux-enabled Server: Access it in your browser or via `curl`
`curl http://localhost:8080` expected output: This is the SELinux-enabled server
- 2-SELinux-disabled Server:
`curl http://localhost:8081` expected output: This is the SELinux-disabled server

Step 4: Simulate an Attack

1. Create a restricted file
`sudo echo "Restricted content" | sudo tee /var/www/restricted.txt`
2. Attempt Access on SELinux-Enabled Server:
`curl http://localhost:8080/../restricted.txt`
Desired Result: SELinux will block access, and you should see an error.
3. Attempt Access on SELinux-Disabled Server:
`curl http://localhost:8081/../restricted.txt`
Desired Result: The file may be accessible because SELinux is not enforcing restrictions.

Step 5: Analyze SELinux Logs

For the SELinux-enabled server, you can check the logs to see the access denial:

```
sudo ausearch -m AVC
```

SELinux will log any attempts to violate its policies.

Conclusion

- **SELinux-Enabled Server:**
 - Access to sensitive files is blocked.
 - Unauthorized actions are logged.
 - The server remains secure even with misconfigurations.
- **SELinux-Disabled Server:**
 - Files outside the web directory may be accessed.
 - Unauthorized actions are not logged.
 - The server is vulnerable to exploitation.